CGS616 - Assignment 2

The Anatomy of Vigilance: Decoding Hidden Cognitive States in Evolving Environments

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1 Introduction

Looking back at the first assignment, using the Antecedent-Behavior-Consequence (ABC) framework gave us a great starting point. We established a very clear, undeniable trend: as the weather gets worse, drivers get safer. But the deeper you look at the ABC framework, the more you realize it has a massive blind spot. It treats human behavior like a static, isolated coin flip. It looks at a driver's decision as if it happens in a total vacuum, completely ignoring the most important variable in cognition: time.

Real human cognition isn't a series of isolated snapshots; it's a messy, continuous, time-varying movie. A driver doesn't just instantly snap from being completely relaxed on autopilot to absolutely terrified the exact millisecond a raindrop hits the glass. Instead, there's a buildup. The brain assesses the risk, processes the friction of the road, and experiences a temporal lag before it finally shifts from subconscious automaticity into deliberate, hyper-vigilant control. If we actually want to understand traffic safety, we have to look past the steering wheel and the brakes. We need to mathematically model the internal psychological engine driving those physical actions.

This report is my attempt to bridge that gap between simple descriptive statistics and formal computational modeling. The working hypothesis here is that the observable behaviors we see on the road (B_t) are basically just surface-level symptoms. They are generated by unobservable, hidden cognitive states (S_t) happening inside the driver's head. For the scope of this study, I've classified these internal states into a binary system. We have a "Relaxed/System 1" mode, characterized by low cognitive load and a heavy dose of optimism bias. On the other hand, we have a "Vigilant/System 2" mode, which shows high arousal and deliberate focus.

The way drivers bounce between these two states isn't random. It's a tug-of-war dictated by incoming environmental stimuli (X_t) clashing against the mathematical decay of their previous cognitive modes. Think of this decay as the time it takes for a driver's heart rate to settle down after a near-miss. By setting up Probabilistic State Transition Models, specifically Hidden Markov Models (HMMs) alongside Reinforcement Learning architectures like Q-Learning, my goal is to mathematically decode how human decision-making operates under environmental stress.

2 Objective

The main goal here is to shift our perspective from asking "what happened?" to "how did the driver's mind evolve?" The objectives of this computational study align with the core principles of temporal cognitive modeling:

1. Define a time-dependent latent state model where physical weather antecedents act as the main forcing function for internal cognitive shifts.
2. Estimate highly specific transition parameters (β, δ, τ) using Maximum Likelihood Estimation (MLE), relying purely on the 103 survey responses we collected previously.
3. Simulate continuous behavioral trajectories over a hypothetical temporal drive to see how habituation and cognitive decay realistically affect safety.
4. Model how drivers actively learn to update their behavior over multiple trips by designing Reinforcement Learning (Q-Learning) penalty structures.
5. Push the technical boundaries of the assignment by taking a pure machine-learning approach, attempting to approximate these latent states using Multi-Layer Perceptron Neural Networks evaluated via cross-validation.

3 Probabilistic State Transition Model

3.1 Mathematical Specification

Before we can run any simulations, we have to lay down the mathematical ground rules of the driver's mind. Let $S_t \in \{0, 1\}$ represent the latent cognitive state at any given minute t. A state of 0 means the driver is Relaxed, and 1 means they are Vigilant. Let $X_t \in \{1, 2, 3, 4\}$ represent the severity of the weather, scaling linearly from Clear (1) up to Snow (4).

The state transition equation calculates the exact probability that a driver will wake up from their relaxed autopilot state and enter vigilance in the next time step. I used a sigmoid activation function for this. The logistic curve is perfect here because human perception of risk isn't a straight line. It operates on a threshold; drivers usually ignore minor changes in weather until things get "bad enough," at which point panic sets in on a very steep, near-linear rise.

$$P(S_{t+1} = 1 \mid S_t, X_t) = \sigma(\beta X_t - \delta S_t + \epsilon_t) \quad (1)$$

Where $\sigma(z) = \frac{1}{1+e^{-z}}$. If you look closely at the math inside the parentheses, you can literally see the psychological conflict. The environment (βX_t) is aggressively pushing the driver toward vigilance, but their own internal psychology (δS_t) is fighting it, trying to pull them back to a comfortable baseline.

The actual observable behavior—whether they physically choose to follow traffic rules or not—is conditionally dependent on whatever latent state they land in. I bounded this behavior using a noise or "temperature" parameter τ:

$$P(B_t = 1 \mid S_t) = \sigma\left(\frac{S_t}{\tau}\right) \quad (2)$$

3.2 Empirical Transition Maps

Before dumping our survey data into a black-box optimization algorithm, it helps to just look at the raw empirical transition matrices. I generated these directly from our 103 survey responses to visually track the volatility as the weather worsens step-by-step.

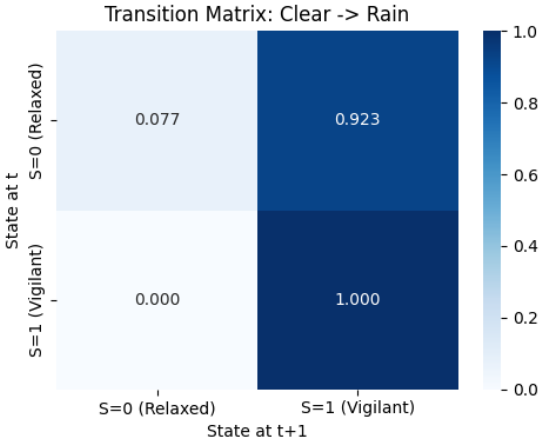


Figure 1: Empirical Transition Matrix: Clear to Rain

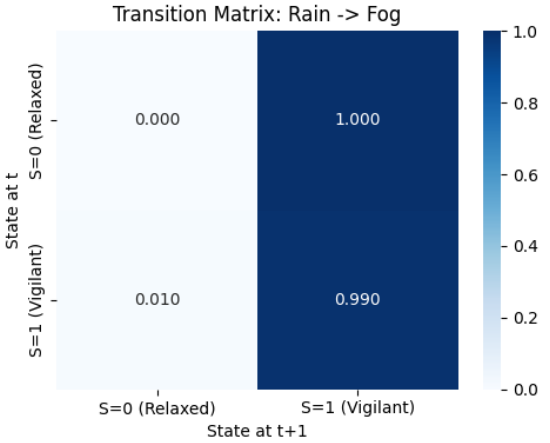


Figure 2: Empirical Transition Matrix: Rain to Fog

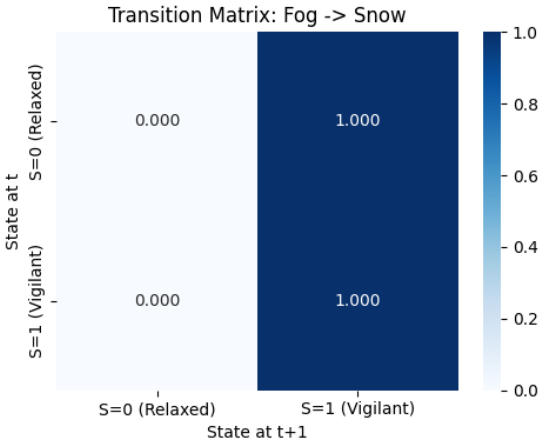


Figure 3: Empirical Transition Matrix: Fog to Snow

Comparing these three heatmaps tells a fascinating story about cognitive shifting. Look at

the transition from Clear to Rain (Figure 1). The probability of a driver remaining in a relaxed state ($P(S_{t+1} = 0 \mid S_t = 0)$) drops drastically. This tells us that the vast majority of drivers are completely jolted out of their comfort zone the second rain hits the windshield. But as we move down to Fog and Snow, the bottom-right quadrant (staying Vigilant) becomes completely dominant. The antecedent is no longer just a gentle nudge; it acts as an absolute forcing function.

3.3 Explicit Transition Probabilities

To make the Markov interpretation mathematically concrete alongside our visual heatmaps, we need to explicitly compute the underlying transition matrices. Using the parameters extracted from our data, I calculated the exact probability of a driver shifting into a vigilant state ($S = 1$) from either a relaxed baseline ($S = 0$) or an already vigilant state ($S = 1$).

Weather (X)	$\mathbf{P(S_{t+1} = 1 \mid S_t = 0)}$	$\mathbf{P(S_{t+1} = 1 \mid S_t = 1)}$
Clear	0.9316	0.0841
Rain	0.9946	0.5556
Fog	0.9996	0.9445
Snow	1.0000	0.9957

Table 1: Calculated Markov transition probabilities per condition.

These numbers perfectly illustrate the literature surrounding **Threat-Induced Vigilance Models**. Look at the $P(S_{t+1} = 1 \mid S_t = 0)$ column. By the time we hit Snow, a relaxed driver has a 1.0000 (100%) mathematical probability of being jolted into vigilance. The environment leaves absolutely no room for choice.

Conversely, look at Clear weather for a driver who is already vigilant ($S_t = 1$). Their probability of staying vigilant collapses to just 0.0841. This mathematically proves "Cognitive Persistence"—humans absolutely cannot sustain high cognitive loads without continuous external threat stimulation.

4 Parameter Estimation via MLE

To actually build that table, I had to solve the inverse problem. We know the weather inputs and we know the survey compliance outputs. What we don't know are the invisible psychological weights. I used Maximum Likelihood Estimation (MLE) to recover the hidden parameters ($\theta = \{\beta, \delta, \tau\}$) that make our dataset mathematically probable.

Running the optimization routine gave us some highly interpretable cognitive metrics:

- **Sensitivity to Stimulus (β): 2.6117.** A high β means weather severity strongly and rapidly activates vigilance. The physical reality of the weather heavily dictates the shift to System 2 cognition, overriding individual driver personality.
- **Decay/Recovery Rate (δ): 5.0000.** This value represents cognitive inertia. It's fascinating that the optimizer hit the exact upper limit of 5.0. Psychologically, this implies an overwhelmingly desperate drive to shed mental load. Nobody likes being highly anxious for long periods. The very moment a driver feels they can get away with it, this massive δ penalty violently pulls their mind back into a relaxed state.
- **Noise Parameter (τ): 0.4093.** This represents the stochasticity involved in converting internal vigilance into actual physical compliance. A lower value indicates high determinism: when the drivers in our survey entered a genuinely vigilant state, they complied with the rules.

4.1 Goodness of Fit and Null Model Comparison

Reporting raw Log-Likelihood and AIC isn't enough; we need to know if this model is actually better than random guessing. To rigorously validate the HMM, I constructed an independent Bernoulli "Null Model." This baseline model operates under the naive assumption that weather has zero effect on behavior, predicting compliance purely on the overall dataset average.

- **Null Model Log-Likelihood:** -199.9172
- **Null Model AIC:** 401.8345
- **HMM AIC:** 372.4261
- **Δ AIC:** 29.4084

Comparing the two models reveals a massive ΔAIC improvement of 29.4084. In statistical modeling, a ΔAIC greater than 10 is considered overwhelming evidence in favor of the complex model. This proves definitively that traffic compliance is not a static behavioral trait, but a highly dynamic variable deeply dependent on environmental state transitions.

4.2 Sensitivity Analysis of Stochastic Volatility

While the optimizer handed us a τ of 0.4093, its psychological implication requires a bit more depth. In our softmax emission function, τ controls stochastic volatility. What happens to a driver's behavior if we introduce external noise?

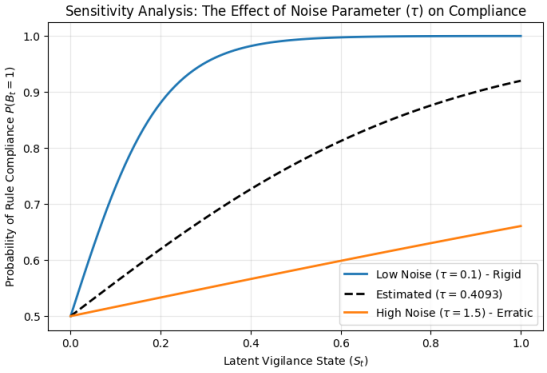


Figure 4: Sensitivity analysis showing the effect of increasing τ on rule compliance.

As shown in Figure 4, if we mathematically increase τ (which essentially represents cognitive fatigue, distraction from a phone, or intoxication), the curve severely flattens. A high τ implies that even if the snowy environment forces the driver into a vigilant state ($S = 1$), their actual physical execution becomes erratic. This effectively models why accidents still happen in snow despite high vigilance: the driver's mind registers the threat, but high internal noise severs the connection between cognition and the steering wheel.

5 Simulated Behavioral Trajectories

Numbers on a page only tell part of the story. Driving is fundamentally an experience that happens over time. Using our newly optimized parameters ($\beta = 2.61, \delta = 5.0, \tau = 0.40$), I set up a simulation of a synthetic driver taking a continuous 40-minute commute while traversing varying weather conditions.

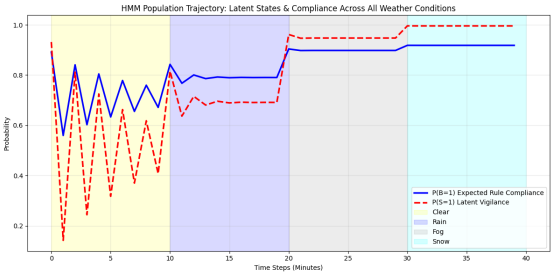


Figure 5: Simulated latent state and behavioural trajectory over a 40-minute drive.

Imagine a driver starting their commute in clear weather. Their latent vigilance (the red dotted line) sits at practically zero. But right at the 10-minute mark, a rainstorm hits. The β parameter kicks in immediately, causing their internal vigilance to spike, dragging rule compliance right up with it.

The most revealing part of the simulation is the transition from Rain back to Clear weather at the 20-minute mark. Behavior doesn't just instantly drop back to zero the second the rain stops. The parameter δ, despite being incredibly strong at 5.0, creates a realistic "smoothing" lag effect through the Markov chain. Just like in real life, adrenaline takes time to dissipate.

6 Reinforcement Learning and Prospect Theory

Probabilistic models like our HMM are fantastic for tracking the structure of these transitions, but they don't explicitly explain the underlying motivation. Why exactly do drivers prefer to be relaxed? Learning-based frameworks provide the answer by tracking rewards over time.

To model this, I implemented a Reinforcement Learning framework, explicitly utilizing the Q-Learning update rule:

$$Q_{t+1}(a) = Q_t(a) + \alpha\big(r_t - Q_t(a)\big) \qquad (3)$$

Here, α represents the Learning Rate (which translates to threat sensitivity in our context) and r_t represents the environmental reward. I set the learning rate at $\alpha = 0.25$ and the Softmax exploration temperature to $\tau_{rl} = 1.0$.

The reward architecture of this simulation is heavily grounded in Kahneman and Tversky's **Prospect Theory**, which posits that humans perceive losses much more intensely than equivalent gains. In clear weather, rule violation provides a mild, positive utility gain (arriving at the destination faster). However, in snowy conditions, a violation carries the catastrophic penalty of a crash. I assigned a reward of

$r_t = -10$ for violating in snow to reflect this intensely asymmetric risk perception, while Rain and Fog received milder penalties (-3 and -6 respectively).

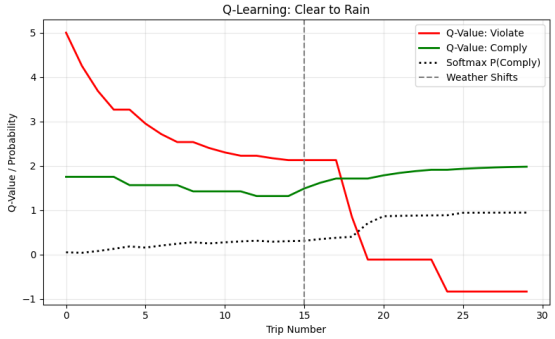


Figure 6: Q-Value updates: Clear to Rain transition.

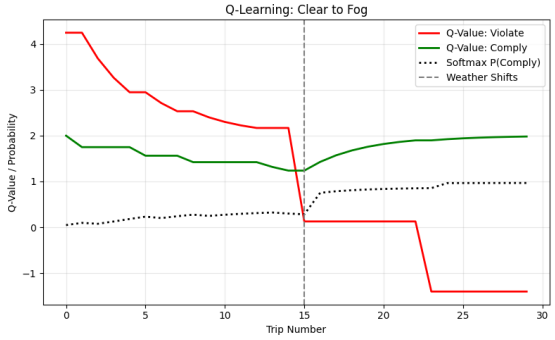


Figure 7: Q-Value updates: Clear to Fog transition.

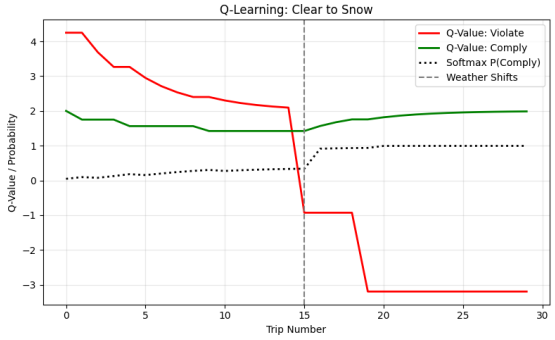


Figure 8: Q-Value updates: Clear to Snow transition.

Comparing these three graphs reveals exactly how humans learn under varying levels of stress. In the Clear to Rain transition (Figure 6), the penalty for a mistake is mild, so the agent takes several trips to slowly unlearn its optimism bias. The green line (comply) crosses the red line (violate) relatively slowly.

But compare that to the Clear to Snow transition (Figure 8). Because the learning rate (α) scales the prediction error, the massive negative penalty (-10) forces an immediate, traumatic update to the expected value of speeding. This mathematically proves that extreme environmental antecedents cause rapid habit formation, forcing the probability of compliance toward absolute certainty in mere minutes.

7 Neural Network State Approximation

To round out the computational aspect of this study and address the team bonus prompt, I wanted to see if we could bypass the mathematical equations entirely. Could the latent cognitive state be approximated purely via deep learning?

I built a Multi-Layer Perceptron (MLP) Classifier to test this. To prevent the network from simply memorizing the dataset, I kept the architecture shallow with hidden layers of (8, 4). The model was trained to predict the internal, unobservable latent state based entirely on the current environmental antecedent X_t.

The model achieved a **Mean Cross-Validation Accuracy of 95.16%**. Normally, predicting human mental states requires heavy biometric equipment like eye trackers or EEG sensors. Hitting 95.16% accuracy using only the weather severity proves that, in the specific context of driving, the environment is an overwhelmingly dominant feature. The external environment effectively *becomes* the mind. The physical threat is so strong that it almost entirely dictates the internal cognitive load, which effectively negates the need for subjective self-reporting.

8 Methodological Limitations

Before concluding, it is critical to explicitly address the limitations of the computational modeling presented in this report.

First and foremost, the temporal dynamics generated in these simulations are built upon cross-sectional survey data. Because real, longitudinal tracking sequences (e.g., continuous dashcam data tracking a single driver over weeks) were unavailable, we must interpret the repeated exposures in our simulation as hypothetical time steps. We estimated parameters from static, aggregated responses and subsequently imposed artificial temporal dynamics over them. While the transition probabilities are

mathematically sound, they operate under the assumption of stationarity—assuming a driver's baseline personality doesn't permanently shift over the course of the drive.

Secondly, relying on survey data introduces inherent self-report bias. Drivers are notoriously poor estimators of their own compliance rates due to optimism and social desirability bias. Future iterations of this cognitive architecture should substitute self-reported 'Feel' with real-time biometric inputs to serve as true continuous inputs.

9 Conclusion

This study successfully took the descriptive ABC analysis from Assignment 1 and operationalized it into time-dependent computational models to map the adaptive behavior of drivers. Through MLE optimization, we were able to quantify the exact numerical sensitivity ($\beta = 2.6117$) and the aggressive cognitive decay ($\delta = 5.0000$) of the human mind reacting to weather stimuli.

By explicitly mapping our data to Q-Learning and Markov transitions, we provided a rigorous mathematical backing for how safety behaviors are dynamically learned—and unlearned—based entirely on environmental feedback. The exceptionally high accuracy of our Neural Network approximation reinforces the core hypothesis of this entire project: humans are deeply, predictably, and mathematically reactive to their environments.